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Review of transportation choice research in Australia: Implications for sustainable urban transport design

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Abstract

In Australia, the transportation sector is the third largest contributor of CO_2-e emissions. In order for Australia to meet the ratified target of 5% below 2000 emissions levels by 2020, a number of changes must be put in place. Sustainable transport design allows for more carbon-friendly travelling alternatives in urban and suburban areas where transportation activities mostly occur. In order for the design to be successfully accepted by the community, a good understanding of people's travel behaviour and preferences of transportation mode is required. This paper offers a review of current transportation research in the context of Australia that spans a number of disciplines including economics, engineering and psychology. The objectives are to assess the key factors that have been empirically shown to affect individual transportation mode choice, identify the shortcoming and gaps in the literature, and recommend how this research can be used to aid in future carbon reduction through better understanding of people's travelling behaviour.

Keywords: Travel demand model; Psychometric scales; Attitude; Choice; Policy.

1. Introduction

Environmentally sustainable transport (EST) is defined broadly by the Organisation for Economic Co-operation and Development (OECD, 2000) as transport that provides access to people, places, goods and services while preserving public health and environmental quality. With specific reference to carbon emissions, EST should limit carbon emissions to only 20% of the emissions level in 1990 by year 2030 and utilize renewable and non-renewable resources at a rate below or equal to their regeneration (OECD, 2002).

Sustainable urban designs can result in possible carbon cuts from the transportation sector. This highlights the need for better understanding of the various underlying factors involved in the decision-making processes and their integrated effects on the transportation choice of individuals in a country.

As part of Australia's ratification of the Kyoto Protocol in 2007, Australia has committed to the development of a Carbon Pollution Reduction Scheme, which legislates the reduction of Australia's emissions to at least 5% below 2000 emissions levels by 2020 (DCC, 2010). At the time of signing the Protocol, Australia's net greenhouse gas emissions was over 800 megatonnes (Mt) of carbon dioxide equivalent (CO_2-e). This figure was 82% higher than the 1990 level. On a per capita basis, Australia's carbon dioxide emissions were 4.5 times the global average (Raupack, 2007). From 2000–07, per capita emissions grew from 18.2 to 19 metric tons, or an increase of 4% (UN, 2010). With a growing population that is expected to reach 62.2 million by 2101 (ABS, 2010), a 5% commitment would appear to be a daunting task.

Australia has 15.7 million registered motor vehicles (ABS, 2009) for a country of around 22 million people (i.e., approximately 720 vehicles for every 1,000 residents). Currently, 15.8 million people live in the five major cities of Australia, which are Sydney, Melbourne, Brisbane, Adelaide and Perth. Australian cities are quite decentralized with distances between cities ranging from 230 to nearly 4,500 kilometres. Figure 1 presents the driving distances between each major city based on estimations by the Roads and Traffic Authority, New South Wales (GoNSW, 2002). The growing population and urban sprawl result in suburbs that are highly automobile-oriented or dependent; this in turn leads to a larger carbon footprint.

The number of kilometres travelled for all registered vehicles in Australia for 2007 was 215 billion kilometres (ABS, 2007). Passenger vehicles accounted for 73.4% of

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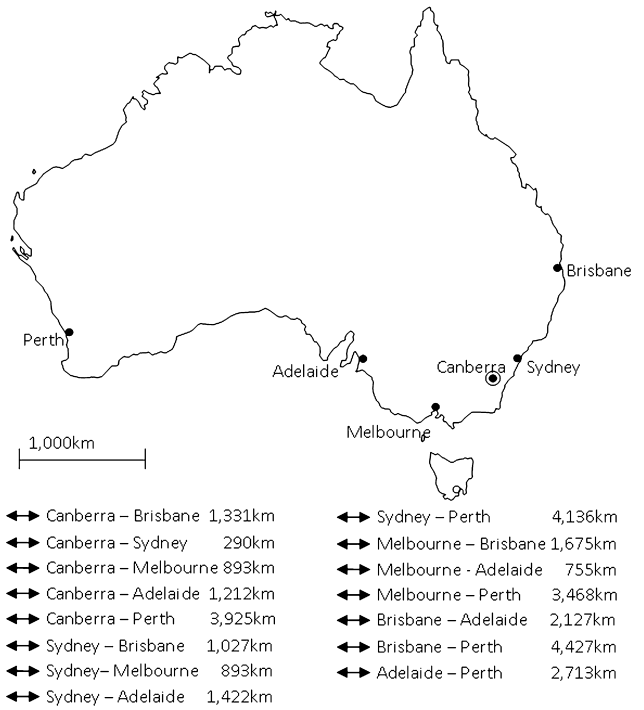

Figure 1. Driving distances between each major city in Australia. *Source*: GoNSW (2002).

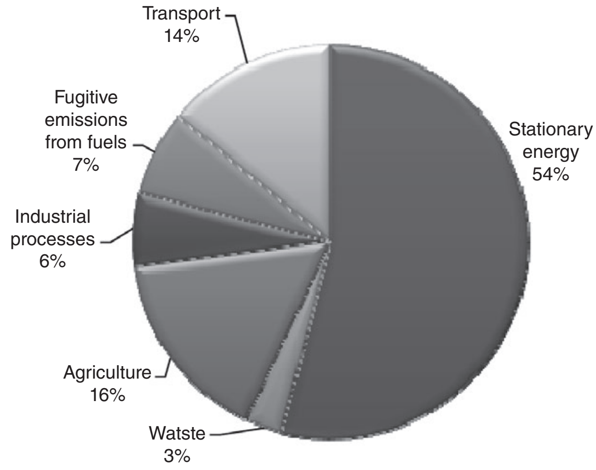

Figure 2. CO_2-e emissions contribution per sector in Australia for 2007. *Source*: DCC (2010).

the total distance travelled of all vehicles in Australia in the 12 months preceding 31 October 2007, accounting for approximately 158 million kilometres. Personal and other use accounted for 51% of that figure, followed by travel to and from work, at 28.7%, and business use, at 20.2% (ABS, 2007).

Fuel consumption for passenger vehicles was 18.1 billion litres in the 12 months preceding 31 October 2007 or approximately 60% of total fuel consumed by all registered motor vehicles in Australia (ABS, 2007). Reflecting the reliance of the Australian society on motor vehicles, the transport sector is the country's third largest contributor of CO_2-e emissions, contributing approximately 14% in total (see Figure 2), following the stationary energy sector at 54%, and the agriculture sector at 16%, respectively (DCC, 2010).

One option for reducing per capita carbon emissions from the transport sector is to encourage people to use fewer carbon or fuel dependent types of transport, in particular, reducing the kilometres driven by personal passenger vehicles. This may be achieved through encouraging the use of more sustainable transport choices such as public transport, walking or cycling. However, changing people's behaviour is difficult when there is no supporting infrastructure, such as better coverage of bus routes or train lines. On the other hand, before designing transportation infrastructure, one should have a good understanding of how people make travelling and transport decisions. It is therefore extremely important to understand population mobility dynamics and the human decision-making processes relating to transport mode choice.

The purpose of this paper is to present a concise review of the literature relating to transport mode choices in Australia in order to: 1) flesh out the key variables that have been found empirically to influence people's transport mode choice; and 2) to identify some priority research areas to fill in the gaps in the current literature. To the best of our knowledge, it is the first such review on Australian transport studies.

In order to identify relevant studies, the following databases have been extensively searched and reviewed: Econlit, Web of Science, Google Scholar and Sciencedirect. The reference lists of the studies found through that search have been used in a snowball search for additional publications about Australian transport choice. Where suitable, studies from outside of Australia have been consulted to further inform understanding.

2. Factors affecting transportation mode choice for Australians

There is a substantial amount of evidence in the literature to suggest that economic, social psychological, physical (e.g. built infrastructure), political and climatic factors impact individual transport mode choice. The main findings from Australia-based studies are presented in Table 1. Signs of individual variables (where available) indicate the direction of their impact on the dependent variable of the study. These factors are discussed in the following subsections.

2.1. Demographic factors

Among the demographic factors, family size, age and gender are identified as the key variables in Australian studies. Family size or number of persons in the household

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Table 1. Summary of recent major studies on the identification of significant variables affecting travel choice in Australia

Study	City / location	Dependent variables	Explanatory variables				
			Demographic	Economic	Physical	Social Psychological	Climate
Wen *et al*. (2010)	Sydney	Driving to work	• Tertiary education • Full-time employment • No. of children in the household >1 • Parent age >=40 yrs	• No. of cars in the household >=2	• Workplace encourages active travel • Convenient public transport close to work • Workplace has shower and change rooms • Convenient parking near workplace • Convenient public transport • Distance to work > 5 Km • Distance to work >10 Km	• Workplace is in a safe area	
Badland and Duncan (2009)	Queensland	Walking or cycling to/from work					• Air pollution [–]
Hume *et al*. (2009)	Melbourne	Walking or cycling by children and adolescents				• Interaction with neighbours [+] • Perception of insufficient traffic lights [–] • Perception of insufficient pedestrian crossings [–]	
Garrard *et al*. 2008)	Melbourne	Female commuter cyclists			• Routes separated from motorized traffic [+]		
Giles-Corti *et al*. (2008)	Western Australia	Probability of walking			• Pedestrian-friendly neighbourhood [+]		
Wen *et al*. (2008a)	Western Sydney	Children being driven to school		• Number of cars in the household [+]	• Distance from home to school [+]	• Mode of parents' travel to work (travel habit) • Parent's attitudes	
Golob and Hensher (2007)	Sydney	Trip chaining behaviour	• Age [–] • Gender, female [–] • Living condition, single [–]	• Household income [+]			
Shannon *et al*. (2006)	Perth	Commuting in an active mode (cycling, walking or public transport) to university					• Bad weather (rain, wind or heat) [–]
Soltani and Allan (2006)	Adelaide	public transport	• Female [–] • Adult living alone [+]	• Household income [+] • Employment (Student) [–] • Car licence [+]			
		Walking / bicycling		• Household income [+]	• Weekday [–] • Route directness (RD) [–]		
		drive alone	• Number of members [–]	• Household income [+]	• Travel time (min) [–] • Number of stops [+] • Car licence [+] • Distance to Central Business District [+] • Employment density [–] • Proximity to shopping [+]		
Collins and Chambers (2005)	Melbourne	Use of public transport to travel to university		• Petrol cost [+] • Parking cost [+] • Travel time [–]			

Notes: [+] Indicates positive impact and [–] indicates negative impact of explanatory variables on dependent variable.
Source: See Iftekhar *et al*. (2010) for greater detail.

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can affect travel choice. Soltani and Allan (2006) have observed a negative relationship between numbers of members in the household with the probability of driving alone. Age is found to negatively affect travelling distance and preference to drive cars. Golob and Hensher (2007) observed that travel demand shifts from car driving to car passenger and then to public transport in complex trip chains after the age of 64, especially for singles and women.

Bell *et al*. (2006) compared 1996 and 2001 Census data on travel-to-work behaviour from employed persons aged 15 years and over to determine prevalence and trends in walking and cycling to work. Responses were compared by state, gender and age group. They observed that in 2001, 3.8% of Australians walked to work and less than 1% cycled. Over 64% travelled to work by car. There was a small decline in both walking (men and women) and cycling (men) between 1996 and 2001. People were more likely to walk or cycle to work if they were male or if they were aged 15 to 24 years.

For children in Australia, the car is becoming the main mode of transport to school. Van der Ploeg *et al*. (2008) provided a long-term trend (between 1971 and 2003) analysis of how children in Sydney travelled to and from school. They observed that the percentage of children aged 5 to 9 that walked to school had decreased from 57.7 to 25.5% between 1971 and 2003. On the other hand, the percentage of children aged 5 to 9 that were driven to school by car had increased from 22.8 to 66.6%. The results for children aged 10 to 14 were similar, walking decreased from 44.2 to 21.1% and car use increased from 12.2 to 47.8% over the study period. There were no major differences between boys and girls. Tranter and Whitelegg (1994) studied the travel behaviour of children in Canberra. They observed that the preferred mode of transport varies with age. For example, the proportion of children travelling by car to school for ages 9, 10 and 11 were 51%, 38% and 39%. Similar figures for cycling were 9%, 13% and 12%; for bus 8%, 9% and 9%; and for walking 32%, 41% and 39%.

In another study in Western Sydney, Wen *et al*. (2008b) examined the factors associated with children being driven to school. They observed that 41% of students travelled by car to or from school for more than five trips per week. Almost a third (32%) of students walked all the way. Only 1% of students rode a bike and 22% used more than one mode of travel. Of those who were driven, 29% lived less than 1 km and a further 18% lived between 1 and 1.5 km from school. Hume *et al*. (2009) observed that the tendency of active commuting (e.g. cycling or walking) changes as children get older.

In summary, between 1971 and 2003, Australian children's mode of travel to and from school has markedly shifted from active (walking) to inactive (car) modes. Increased travelling distance to school does not necessarily translate to a higher percentage of car use. Overall, the factors associated with car travel for children were the mode of travel to work used by parents, parent's attitudes and the number of cars in the household. As for adults, a similar decline in cycling and walking trends is also observed.

2.2. Economic factors

Economic factors that have been found to significantly affect transport mode choice include household income, transportation fare and employment status. The type of employment can have a significant effect on the mode choice. The largest numbers of commuters are likely to be found in suburban employment sub-centres, which often have a limited supply of public transport from areas other than the city centre. For these suburban sub-centres, it is frequently the case that no travel mode is available other than the private car. Living in an area with higher employment density reduces the propensity of driving a car due to the limitations of parking or traffic congestion (Vega and Reynolds-Feighan, 2009).

Soltani and Allan (2006) observed that in Adelaide as income increases, the likelihood of choosing to travel as a sole driver increases. Interestingly, the relationship between cycling and income is mixed and is correlated to a number of other factors, such as safety of storage facility and healthy lifestyle choices (Wigan, 1984).

Financial cost, such as travel fares, parking fees and petrol prices, can also influence mode choice, especially for low-income people. Among university students in Melbourne, Collins and Chambers (2005) found that 50% of public transport commuters would drive to university if the cost of driving were lower. On the other hand, 70% of car commuters would use public transport if it were faster than car travel. Hensher and King (2001) estimated that a 1% increase in hourly parking fees results in a 0.29% increase in the use of public transport into the Sydney Central Business district (CBD) and a 0.36% increase in commuters driving to free parking located beyond the fringe of the CBD.

Price elasticity estimates provide us a clear picture of the direction and extent of influence of price changes on travel behaviour, which is usually reported in percentage change. Breunig and Gisz (2009) found that the price elasticities of petrol demand for Australia are -0.13 (short-run) and -0.20 (long-run). This suggests that in the short run a 1% increase in petrol price would cause the demand for petrol to drop by 0.13%. Similarly, a 1% increase in petrol price would reduce demand for petrol by 0.20% in the long run. Hensher (2008a) has provided a meta-analysis of the prices and services of public transport and calculated the mean elasticity for fares to be -0.40, suggesting that an in increment of fares by 1% would reduce the demand for public transport by 0.4%. This figure is close to the -0.38 reported in another meta-analysis by Holmgren (2007).

Hensher and Stanley (2009) used a simulation model TRESIS to assess the influence of higher fuel prices on short-run and long-run passenger travel activity in

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Melbourne. They evaluated the effect of a petrol price increase from AU$2[1] to AU$10 over the period 2009–17. Using 2007 as the base year (AU$1.25 and AU$1.40 for petrol and diesel, respectively), they observed prices as high as AU$2 by mid-2009, resulting in a 59.5% increase in car operating cost per kilometre, which in turn resulted in a 6.5% reduction in overall vehicle kilometres and a 6.8% reduction in car CO_2 emissions. By 2017, the progressive AU$1 annual increase in fuel costs would cut annual passenger vehicle kilometres by 25%, and CO_2 emissions from cars by almost 30%.

In summary, the price elasticity of the demand for petrol in Australia is relatively low in the short run (-0.13) and slightly higher in the long run (-0.20). Public transport commuters are more sensitive to fares as shown by Hensher (2008b) and Holmgren (2007).

2.3. *Physical or built environment factors*

Among the physical or built environment factors, travelling distance, urban design, and infrastructure facilities are identified as key variables in Australian studies. Distance, either commuting distance or the distance between activities, is almost always considered as a factor when investigating an individual's transport mode choice. An increase in the travel distance results in an increase in the time and effort needed for travelling (Truong and Hensher, 1985). Soltani and Allan (2006) reported variation in travel behaviour in four suburban residential areas in metropolitan Adelaide. They observed that areas with a higher average commuting distance also have a higher percentage of car use for commuting and rely less on walking.

Similarly, Burke and Brown (2007) observed that in Brisbane the median distance people walk from home to all other places using the walk mode was only 780 metres; from home to public transport stops, 600 metres; and from public transport stops to end destinations, 470 metres. However, cyclists are responsive to dedicated bicycle infrastructure (Heinen *et al*., 2010). For example, female commuter cyclists in Melbourne, Australia prefer to use routes with maximum separation from motorized traffic (Garrard *et al*., 2008).

Based on a survey of Western Australian commuters, Giles-Corti *et al*. (2008) have observed that more people walked for recreation and transport within the neighbourhood (52.6% and 36.1%, respectively) than outside the neighbourhood (17.7% and 13.2%, respectively). Notably, only 20% of average total duration of walking (128 minutes per week) was transport related and within the neighbourhood. When asked about the factors which influenced their choice of housing development, participants identified a new neighbourhood's walkability as an important factor. Similar observations about the importance of walkability were found by Owen *et al*. (2007).

Studies from outside Australia also found that an increase in trip distance results in cycling having a much lower share in mode choice and for commuting. For example, Kahn and Morris (2009) observed that distance to the CBD has a significant negative impact on the probability of commuting to work by using bike and walking. Cao *et al*. (2009) found that neighbourhood characteristics (the built environment and its perception) have a significant influence on travel behaviour. The urban characteristics are often influenced by the land use pattern, public traffic facilities and spatial attributes of residential environment (Cao *et al*. 2009). Schwanen *et al*. (2001) analysed the impact of monocentric and policentric urban structures on mode choice and travel distances for different travel purposes in the Netherlands. They have observed that sparse distribution of urban infrastructures encourages driving and discourages the use of public transport as well as cycling and walking.

In summary, there are many facets of the built environment that could encourage the use of more sustainable transport mode choices (Mokhtarian and Cao, 2008). These include the level of accessibility of bus stops, walkways and bicycle paths; the level of safety, particularly for pedestrians and cyclists; and the ease of use (Cervero, 2002). The link between sustainable urban design and sustainable transport choice is clear. A well-designed neighbourhood with convenient bicycle and walking paths that are well connected to business facilities and public transport terminals should be able to encourage less dependency on private car use in the future, regardless of whether or not the suburb is located further away from the central business district (Chen *et al*. 2008).

2.4. *Climate factors*

The analysis from the Stern review (Stern *et al*., 2006) suggests that global mean temperatures could rise by 2 to 5 °C if greenhouse gas concentrations double compared to their pre-industrial level. According to the Intergovernmental Panel on Climate Change (IPCC) report, the projected annual average temperature around the inland coastal region of Australia would increase between 0.3 to 2.7 °C by 2050, while average annual rainfall would drop between -40% and 0% (Hennessy *et al*., 2007). Extreme weather events such as cyclones, heatwaves and floods not only affect lives, but also businesses and the transport industry. With future temperatures forecast to rise and extreme weather events to become more frequent due to climate change, understanding how climate variables affect transportation choices and people's sensitivity to changes in the climate becomes a crucial factor.

Temperature, rainfall and air quality are the climate factors identified as significant variables affecting transport mode choices in Australia. For example, Nankervis (1999)

[1] Current currency exchange rate is 1 AUD = 0.96 USD.

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found that more people cycle in summer and autumn compared with winter and spring, because cold temperature is considered as more unpleasant than hot temperatures for cycling. Keay and Simmonds (2005) found an overall reduction in traffic volume in Melbourne of 1.35% on wet days in winter and of 2.11% on wet days in spring. Their results also showed an overall volume reduction of 2% to 3% for 2 millimetres to 10 millimetres of rain during the daytime, with reductions in the spring somewhat larger than those in winter.

Similarly, based on an online survey of students from the University of Western Australia, Perth, Shannon *et al*. (2006) observed that bad weather (rain, wind or heat) prevented the students from commuting in an active mode (cycling, walking or public transport). In another study, Badland and Duncan (2009) found that 13% of a sample of commuters in Queensland considered air pollution a major barrier to walking or cycling to and from work.

To summarize, while the effects of weather conditions and air quality on the transport sector are evident, the long-term impact of weather pattern changes on people's transportation choices is unclear. In particular, it is unclear as to whether an increase in temperature and reduction in rainfall would shift people's transport behaviour towards more sustainable options, especially in hot Australian climates. The important finding here is that the current transport infrastructure is susceptible to extreme weather events, which in turn impacts commuters' choices.

2.5. Social and psychological factors

A number of theories in the psychology literature have been applied in transportation choice research in order to explain the link between people's attitude and their behaviour; one of these theories is the Theory of Planned Behaviour (TPB). The TPB suggests that attitude, subjective norms and perceived behavioural control can account for a considerable part of the observed variance in observed behaviours (Ajzen, 1991). However, other schools of thought suggest that people's travel behaviours can be habitual (see Verplanken *et al*., 1994 and Bamberg and Schmidt, 1998) and may not always be preceded by deliberation, implying that incorporating a measure of habit may help improve the predictive capability of the attitude-behaviour model (Anable, 2005). There is also empirical evidence suggesting that people's travel behaviour follows a stage or cycle (Bamberg, 2007) where an individual goes through stages of pre-contemplation, contemplation, preparation, action and maintenance. This is based on the Transtheoretical model proposed by Prochaska and DiClemente (1984).

In actuality, literature on psychological factors affecting transport mode choice is abundant, but only a few empirical studies have been conducted in Australia. We recommend a review by TravelSmart Victoria (2010) as a source of information for psychology theories related to transportation behaviour in Australia. Here we focus on reviewing empirical studies in Australia where psychological factors are used as predictors of transportation mode choice. Hume *et al*. (2009) focused on the changes in active commuting as children get older. They analysed predictors of increases in children's and adolescents' active commuting (walking or cycling) to and from school over a two-year period (2001-04) in Melbourne. They observed that children whose parents knew many people in their neighbourhood were more likely to increase their active commuting compared with other children, with an odds ratio (OR) of between 1.2 and 5.9. Adolescents whose parents perceived that there were insufficient traffic lights and pedestrian crossings in their neighbourhood were less likely to increase their active commuting over 2 years, with an OR of between 0.2 and 0.8. By contrast, adolescents whose parents were satisfied with the number of pedestrian crossings were more likely to increase their active commuting compared with other adolescents, with an OR of between 1.1 and 5.4.

In a study in Western Sydney, Wen *et al*. (2008b) examined the factors associated with children being driven to school. They observed that the parents' mode of travel to work and parents' attitudes were the two significant social psychological factors associated with car travel of children going to school. If the parents believe that their child has the road safety skills needed to walk to school, the OR of the child travelling by car drops from 1 to 0.63.

More often than not, social psychological factors work interactively with or are influenced by other factors (Gärling *et al*., 2002; Gärling and Charles, 2004, Gärling, 2009). For example, based on a survey in six capital cities in Australia in 1994, Golob and Hensher (1998) found that individuals with a strong environmental commitment were more likely typified by young females, from smaller households with fewer cars, having a higher household income and a higher level of education. Conversely, women were more likely to view the car as a status symbol, and tended towards solo driving. Commuters who used public transport were more likely to support policies aimed at reducing greenhouse gas emissions.

To summarize, the social psychology literature stresses the importance of individual subjectivity. For example, safety is measured in terms of the individual's perceived safety instead of actual safety, such as the probability of having an accident. Tversky and Kahneman (1981) argued that people's preferences can be affected by how risk is framed. In other words, at times, information on people's perception can be more important for predicting people's choices than actual knowledge. Modelling people's preferences without verifying the motives behind their values will cause failure in the representation of public opinion and can have significant policy implications (Spash *et al*., 2009).

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3. Measures to promote sustainable transport choice

Recognizing the importance of the different factors affecting transport choices, governments often try to modify them through policy. In the relevant literature, different types of transport policy measures are identified, such as legal policies, economic policies, measures changing the physical context, and informational and educational measures. A distinction is often made between structural or hard measures (i.e., measures intending to alter the individual's context) and psychological or soft measures (i.e., measures aiming to increase awareness and knowledge). Alternatively, policies are often separated into push measures (e.g., prohibiting car use in city centres, raising the tax on fossil fuels, introducing road user charging schemes), and pull measures aiming to improve alternative travel options (e.g., improving public transport, improving the facilities for cycling or walking, introducing park-and-ride schemes and provision of information). The distinction between push and pull measures is particularly important for acceptability, since the public generally assumes push measures to be ineffective, unfair and unacceptable, while considering pull measures to be effective, fair and acceptable (Eriksson *et al*., 2008).

Australia has experienced a number of policies aimed at influencing travel behaviours. James (2002) discusses the success of a low-cost, voluntary behaviour change programme, TravelSmart, implemented in Perth. It was observed that the TravelSmart programme was able to achieve a modal shift and provide a very cost-effective alternative to building expensive road infrastructure solutions (Goulias *et al*., 2002). Rose and Ampt (2001) discussed an awareness programme, Travel Blending(R), that was introduced in Sydney and Adelaide.

It is observed that the public generally prefer "soft" measures. Tay (2005) assessed the performance of the drink-driving enforcement and publicity campaigns in Victoria and found that the campaigns were effective in reducing serious crashes during times of high alcohol consumption. However, studies from other countries indicate that successes of such programmes are directly linked with other factors (Santos *et al*., 2010). Brownstone and Train (1998), based on a stated-preference survey, observed that households who placed relatively little importance on price had a greater tendency to buy an expensive large car and did not react to price increases as readily as households giving average or higher importance to price. Adrián Saldarriaga-Isaza and Vergara (2009) studied the impact of a cash subsidy programme implemented to encourage the use of natural gas in vehicles in the Aburrá Valley in Colombia. They observed that a large part of owners who switched to natural gas would have done so anyway without the subsidy. Ben-Elia and Ettema (2009), in another study in the Netherlands, studied the impact of reward schemes to avoid peak hour driving. They observed that participation in the programme was linked to working time flexibility, personal motivations and household conditions.

The long-term impacts of different policy measures are often overlooked in existing studies. However, long-term and short-term impacts can differ markedly. In Copenhagen, Thøgersen (2009) studied the impact of a field trial where a travel card was provided to car owners for free for a month. They observed that provision of the free travel cards led to a significant increase in commuting by public transport. Although the effect became weaker after the programme expired, they still found a positive effect five months later. Cairns *et al*. (2002) review the impact of several field trials to reduce car use in Europe. Similar types of studies should be conducted in Australia to understand the long-term impact of government policy or business strategies on people's travel behaviour (Stopher, 2004).

Australia also lacks comprehensive studies assessing different policy measures in light of sustainable transport systems principles. Hensher (2008b) studied the potential impact of several policy options regarding travel behaviour with a view to reducing CO_2. Using the TRESIS model, he observed that imposing a 10 cents/km variable user charge on the main road network would reduce car trips by 11%. A 40 cents/kg carbon charge would increase train and bus use by 22% and 16%, respectively (Hensher, 2008b). Policies to increase fuel efficiency would increase car use because of the reduced cost of motoring. Other policies, such as doubling bus frequencies and reduction in rail and bus fares, would also increase public transport use.

4. Limitations of existing research for Australia

Based on our review of transport studies in Australia, we have identified four main limitations or knowledge gaps in transport research.

First, among the economics variables, there is a lack of studies that investigate the impact of employment status and conditions (such as the use of company cars and parking subsidies) on travel behaviour. There is also a lack of understanding of the impact of fare changes on the usage or patronage of public transport. Despite efforts by Hensher and Stanley to estimate a wide range of elasticity responses in the TRESIS model, the question of "What are the elasticities, based on observed actual behaviour?" still remains. Nonetheless, estimates by Hensher and Stanley (2009) are good indicators of what needs to be researched in the future. Emphasis for future studies should be placed on understanding the degree of substitution between modes when there are changes in petrol prices, travel fares and parking fees, as this information would be most critical in terms of understanding how economic tools could be used to induce behavioural changes.

Second, to the authors' knowledge, there has been no published literature about Australia that estimates the

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impact of natural landscape and climate factors on modal choice. The importance of understanding this relationship is that in contrast to motorized transport, choice of non-motorized transport (such as cycling and walking) is strongly determined by the natural environment, with regard to landscape, slope, weather and climate (Müller *et al*., 2008). There is no clear record as to how travel behaviour changes during extreme weather events, such as floods and heatwaves. Heatwaves are very high summer temperatures (around 40 °C or more) that last for several days. Lately, Australia has been experiencing prolonged heatwaves. Apart from damages and disruptions to the road and rail networks, heatwaves have accounted for more deaths in Australia than any other climatic event (GoA, 2010). With the increasing frequency of heatwaves linked to climate change (UN, 2009), it is more important than ever to understand the impact of such weather conditions on people's transportation choices.

Third, most of the past existing research considers the effects of demographic, economic, social psychological, environmental, physical and climate factors in isolation. The "explanatory variables" heading in Table 1 categorizes the five factors that have been found to affect Australian transport choices. Notice that none of the studies listed span across all five factors. Thus, the combined or interaction effects of factors on mode choices have not been investigated. One exception is a study by Golob and Hensher (1998), where gender (demographic) and environmental attitude (social psychology) were linked to transport mode choice. What is more frequently investigated is the interaction effects of two demographic variables such as age and gender (see Golob and Hensher, 2007).

Finally, the purpose of most transportation choice studies in Australia is to understand how behaviour changes when one or more of the factors affecting mode choice changes. This information is normally used to inform how service quality can be improved in order to increase the level of competitiveness of one mode over another. However, there is little understanding as to how destination choice and frequency of visit affect transportation behaviour. For example, people who choose to commute to work by car five days a week may often cycle to the gym on the weekend so that they can incorporate more exercise into their leisure activities. Overall, the literature on leisure or recreational transportation activities is limited in Australia. There is also limited understanding of transportation behaviour for multi-destination trips.

5. Future research direction

There are many ways in which transport mode choice could be modelled, but in the context of understanding carbon emissions in order to promote sustainable transportation design in the future, here are several recommendations for the direction of future research.

5.1. Integration of social psychology into economics

Awareness and acceptance of the interaction between social psychological factors and other factors affecting transport mode choice have led to the development of a methodology for incorporating psychometric data and subjective ratings of attributes into the discrete transport mode choice modelling framework (Walker, 2001; Morikawa *et al*., 2002). The dashed line in Figure 3 shows the traditional decision-making framework based on the utility maximization theory in economics. An individual's decision is influenced by situational, contextual and environmental constraints as well as mode cost, departure and arrival time and number of modes (De Palma and Rochat, 1999). Because an individual's mode preference is based on utility, it cannot be directly observed; however, it is revealed when an individual makes a decision on transport mode. The framework proposed by Walker (2001) and Morikawa *et al*. (2002) allows for the integration of latent social psychological attributes to not only affect utility but also to interact with other factors. The outcome is a model that better represents people's preferences and decision-making complexities.

5.2. Activity-based model

The demand for travel stems from the demand for activities (Bowman and Ben-Akiva, 2001). Most of the transportation studies in Australia are focused on day-to-day commuting activities with a single destination in mind, such as work or school. In reality, people travel for many other reasons and some activities could be as frequent as travelling to work. Some activities are incorporated into other activities, such as stopping at the bank on the way to the supermarket to minimize travelling cost and time. This is called trip-chaining and is often evaluated in tour-based models. From a policy perspective, developing sustainable transportation systems requires an understanding of activity-based transportation behaviour and the drivers of people's route and mode choice where trip-chaining is involved (Currie and Delbosc, 2010).

5.3. A generative approach to transportation choice model

In many real world situations, drivers of transport mode choice interact with each other in multiple dimensions with potential lagged effect and hysteresis. Traditional economic and social analysis tools used in transport research may not be sufficient for studying such phenomena. A generative approach, such as multi-agent systems, could be useful in such situations since it uses an artificial society of programmed agents to simulate the behaviour of a society. Different objective functions, such as individual utility maximization, joint household decisions and life-cycle events could be used to model agents' behaviour. For

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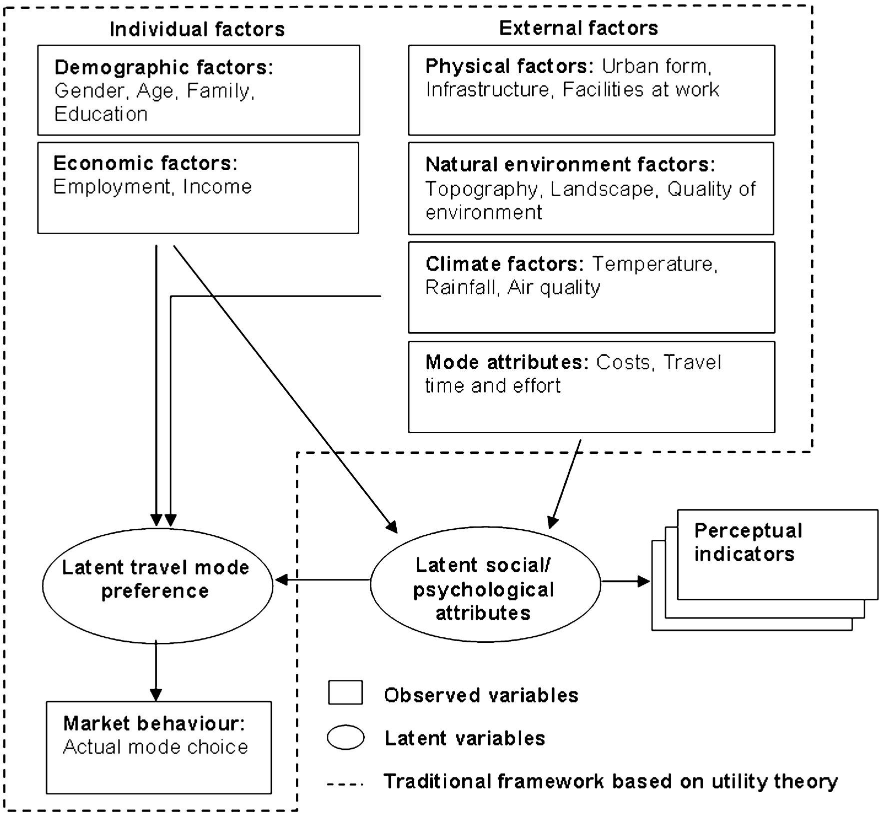

Figure 3. Transport choice model with latent social psychological factors.

example, Miller *et al.* (2005) modelled people's transportation mode choice for activity-based, chain-based and trip-based travel decisions in an agent based modelling framework. Agent-based approaches are inexpensive and can be used to simulate policy measures to study their effect on people's transport behaviour (Epstein, 2006).

6. Conclusion

In this paper, we have reviewed available studies of transport mode choices in Australia to identify the main variables having a significant influence on people's transport mode choice. We have made a qualitative assessment of these factors and indentified some research gaps. In the face of the challenge to develop a sustainable transport system in a changing climate, it is important that researchers in Australia focus on integrating key variables of transport choice in their studies. These variables span across a number of disciplines, including social, economic, psychological, environmental, and climate sciences. Past research indicates that each discipline has been predicting transportation choice behaviour in isolation from the others. In contrast, the review suggests that people's decision-making processes are complex and that attitudinal factors may have as much impact on people's choices as economic and environmental factors. In the absence of a proper appreciation of an integrated framework, there is a high likelihood that predictions of people's choice of travel mode would be based on partial information and be, by consequence, inaccurate. The cost of relying on inaccurate predictions to develop transport policies could be significant, particularly as the Australian Government is trying to curb carbon emissions through massive investments in future sustainable urban design. We expect this paper to be a starting point for future researchers interested in modelling the complexities of transport mode choice. We recommend that a meta-analysis of the studies reviewed in this paper be carried out with a view to quantifying the impact of their key variables.

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